Report 8: The Complete Chain
Bits → DD Grid → Waveform → Noise → Bits

EE451 — Zak Transform / Delay–Doppler Communication
Simulation S8

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1 What this report is about

- Reports 1–7 studied the pieces. Here they are connected into one transmit–receive chain over an AWGN (noise-only) channel, as in the slides' transmitter block diagram.
- Payload: a short speech-like signal, turned into bits by an 8-bit PCM coder, sent through the OTFS chain and rebuilt at the receiver.
- Questions: does the chain give the textbook bit error rate (BER)? What do the different pulses cost? What does a simple equaliser fix?

2 The concept in brief

- **The chain.**

$$\text{bits} \to \text{16QAM} \to X[m,n] \xrightarrow{\text{IZak}} s[a] \xrightarrow{+\text{noise}} r[a] \xrightarrow{\text{Zak}} Y[m,n] \to \text{equaliser} \to \text{bits}.$$

- **Mapping bits to symbols.** Four bits $b_0b_1b_2b_3$ give one 16QAM point. b_0, b_1 choose the sign of the I and Q parts, b_2, b_3 choose the amplitude (1 or 3). The points are scaled to unit average energy. The labelling is Gray: neighbouring points differ in one bit, so a small error usually costs only one bit.
- **Noise and Eb/N0.** A 16QAM symbol carries 4 bits, so $E_s/N_0 = 4\,E_b/N_0$. The Zak transform adds N noisy samples into every DD cell, so we set the per-sample noise variance to $\sigma^2 = 1/(N\,E_s/N_0)$; then every DD cell sees exactly E_s/N_0.
- **Reference BER.** For Gray 16QAM over AWGN,

$$\text{BER} = \tfrac{3}{4}Q(x) + \tfrac{1}{2}Q(3x) - \tfrac{1}{4}Q(5x), \qquad x = \sqrt{0.8\,E_b/N_0}.$$

  If the pulse is orthogonal (Report 4), OTFS should reproduce this curve.
- **What the pulse does (Report 4).** Without noise, $Y = HX$. For the rectangular and orthogonalised pulses $H \approx I$. For Gauss×sinc $H \neq I$: each symbol is polluted by its neighbours along delay (self-interference, rms $\varepsilon = 0.11$ here).
- **Equalisers.** We know H (it is the pulse's own overlap pattern), so the receiver can undo it: zero-forcing $\hat{X} = H^{-1}Y$, or MMSE $\hat{X} = (H + \frac{N_0}{E_s}I)^{-1}Y$, which backs off when noise is strong.
- **From bit errors to speech.** An error in a high-order PCM bit changes the sample a lot; one in a low-order bit hardly matters. So speech quality (SQNR) falls faster than BER suggests.

One-line memory aid: with an orthogonal pulse the OTFS chain behaves like plain 16QAM over AWGN; with Gauss×sinc the self-interference ($\varepsilon \approx 0.11$) adds to the noise and creates a BER floor, which an H-based equaliser removes.

3 What was simulated

- Grid M=32, N=16, D=4, 65-tap pulses: 512 symbols (2048 bits) per frame.
- Payload: a synthetic speech-like signal (a harmonic tone with gliding pitch and syllable-like amplitude modulation, 8192 samples at 8 kHz), coded with 8-bit PCM into 65 536 bits, i.e. 32 frames.
- The rectangular pulse for the speech experiment, at $E_b/N_0 = 4, 8, 12, 16$ dB (10 repetitions each).
- A BER sweep, $E_b/N_0 = 0$–12 dB, for rect, Gauss×sinc (without and with MMSE) and orthogonalised pulses; each point stops at 200 errors or 600 frames.
- A noise-free test, and the eigenvalues of H.

4 Results and discussion

4.1 The 16QAM mapper

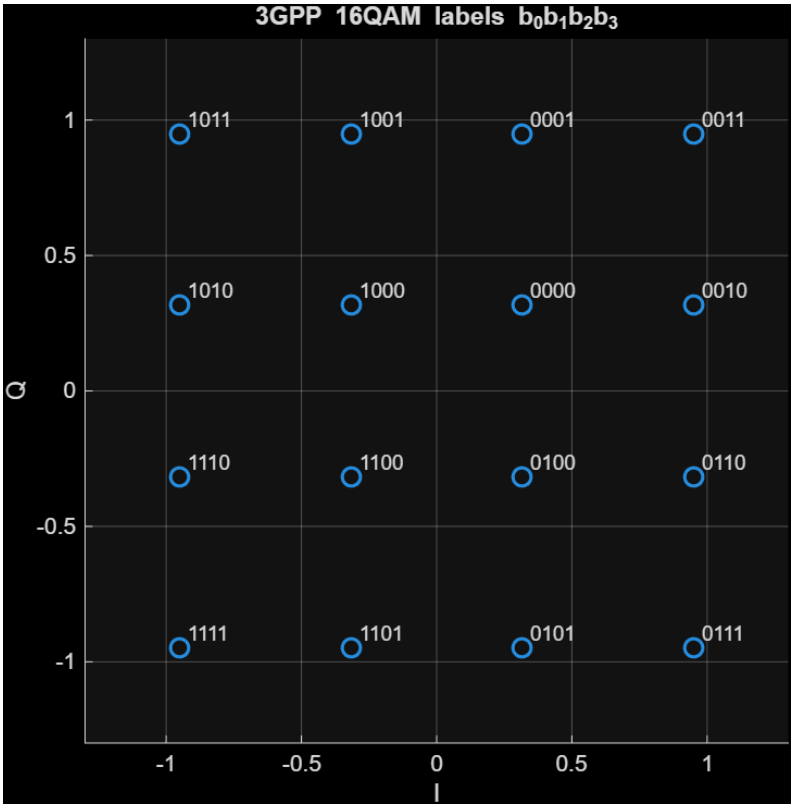


Figure 1: 16QAM points with their bit labels $b_0b_1b_2b_3$.

- **Expected:** 16 distinct points of unit mean energy, with Gray labelling.
- **Observed:** 16 distinct points, mean energy exactly 1.0000, demapping reverses mapping, and every pair of nearest neighbours differs in exactly one bit. In the plot, the first two bits set the quadrant (b_0: left or right; b_1: up or down) and the last two bits the amplitude.
- The mapper implements the 3GPP formula (clause 5.1.4) as we remember it; it has not been checked against the standard's text (see the appendix).

4.2 No noise: bit-exact

- **Expected:** with the rectangular pulse and no noise the chain should return every bit.

- **Observed:** the BER is 0 and the bit stream is identical for all 65 536 bits ("Level-0 loopback"). The 8-bit PCM itself limits the speech quality to an SQNR of 42.12 dB, close to the $\approx$ 42 dB expected from the quantiser step 2/255 and a signal rms of 0.285.

4.3 The received symbols, and the speech

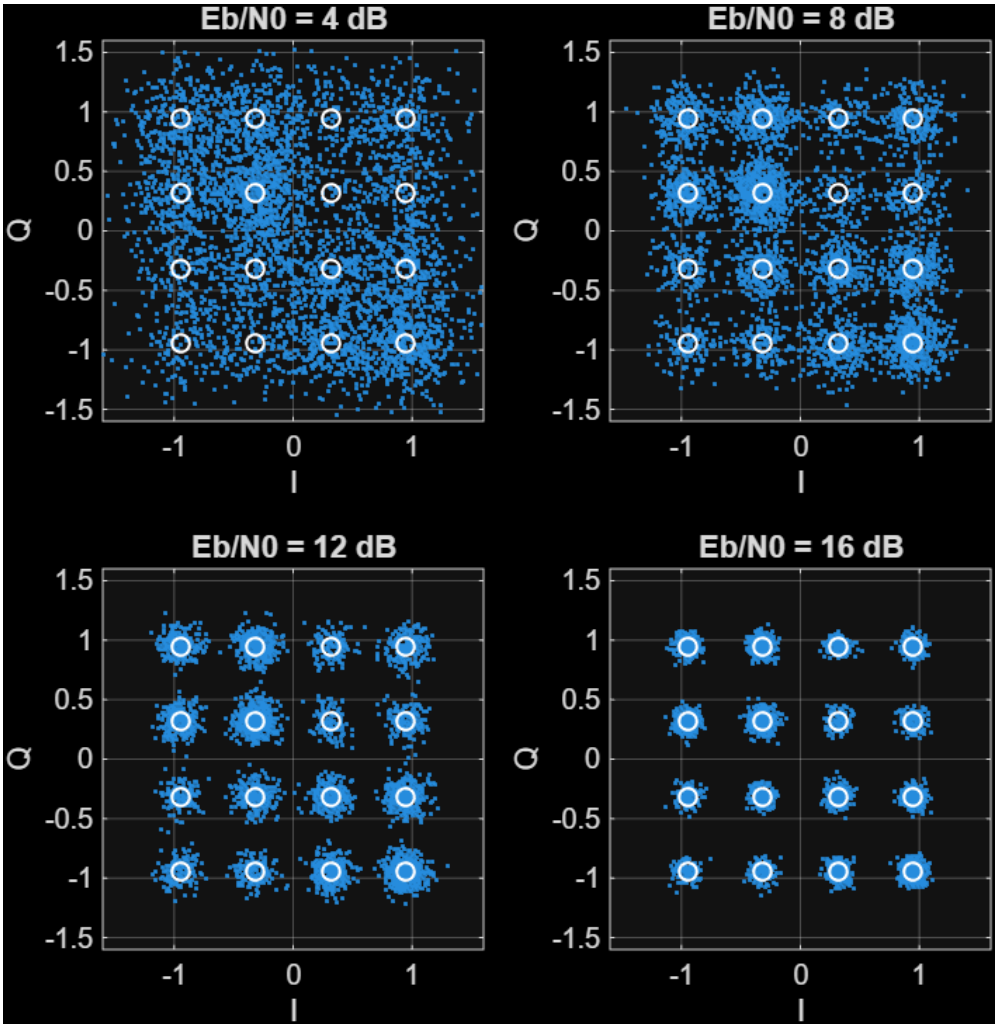


Figure 2: Received DD symbols after the Zak transform (rect pulse, 8 frames); white circles: transmitted 16QAM points.

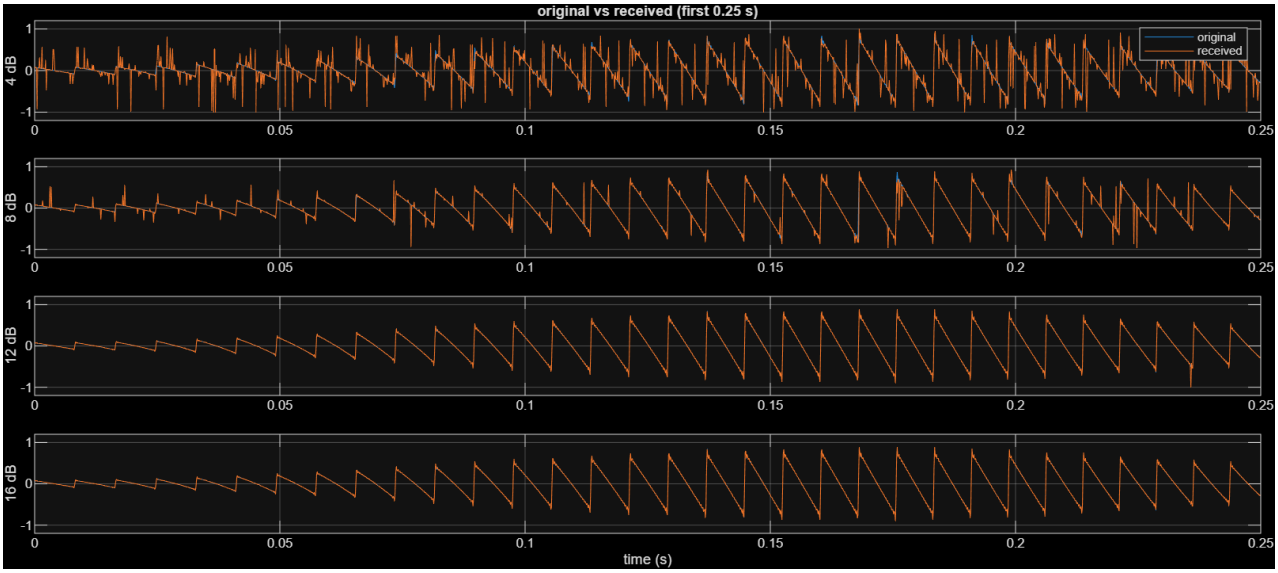


Figure 3: Original (blue) and received (orange) speech for the first 0.25 s.

Table 1: Rect pulse over AWGN, 10 repetitions of the 65 536-bit speech.

E_b/N_0 (dB)	BER (sim)	BER (theory)	SQNR mean (dB)	min	max
4	5.85×10^{-2}	5.86×10^{-2}	1.73	1.41	2.08
8	9.32×10^{-3}	9.25×10^{-3}	9.63	9.29	10.08
12	1.37×10^{-4}	1.39×10^{-4}	28.29	22.91	37.33
16	0	6.25×10^{-9}	42.12	42.12	42.12

- **Expected:** the clouds shrink as E_b/N_0 grows; the speech improves accordingly; the BER follows the 16QAM theory.
- **Observed:** at 4 dB the clouds overlap and the speech is full of spikes; at 8 dB the 16 clusters are visible and spikes are rare; at 12 dB the clusters are tight and only isolated glitches remain; at 16 dB the received speech is indistinguishable from the PCM version (42.12 dB, the quantisation limit). The simulated BER agrees with theory at every point; at 16 dB the theory predicts 6×10^{-9}, i.e. no error is expected in the 6.5×10^5 transmitted bits.
- The SQNR at 12 dB varies widely between runs (22.9–37.3 dB), although the BER is about the same: it depends on whether an error happens to hit a high-order PCM bit.
- Takeaway: the full chain works end to end, and the BER with the rectangular pulse is the textbook value.

4.4 BER with the different pulses

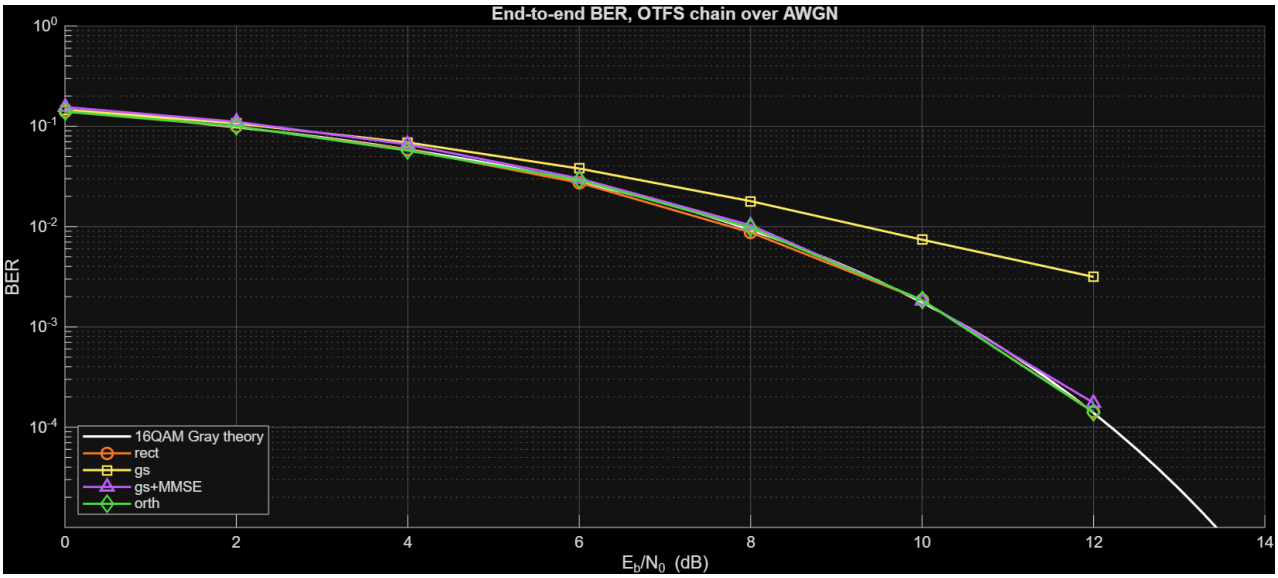


Figure 4: BER against E_b/N_0. White line: 16QAM Gray theory.

Table 2: BER against E_b/N_0 (stop at 200 errors or 600 frames, so about $\pm 7\,\%$ statistical scatter).

E_b/N_0 (dB)	theory	rect	Gauss×sinc	Gauss×sinc + MMSE	orthogonalised
0	1.41×10^{-1}	1.40×10^{-1}	1.46×10^{-1}	1.55×10^{-1}	1.41×10^{-1}
2	9.77×10^{-2}	9.96×10^{-2}	1.07×10^{-1}	1.11×10^{-1}	1.00×10^{-1}
4	5.86×10^{-2}	5.85×10^{-2}	6.88×10^{-2}	6.58×10^{-2}	5.72×10^{-2}
6	2.79×10^{-2}	2.74×10^{-2}	3.79×10^{-2}	3.02×10^{-2}	2.92×10^{-2}
8	9.25×10^{-3}	8.74×10^{-3}	1.79×10^{-2}	1.03×10^{-2}	9.72×10^{-3}
10	1.75×10^{-3}	1.84×10^{-3}	7.40×10^{-3}	1.83×10^{-3}	1.85×10^{-3}
12	1.39×10^{-4}	1.42×10^{-4}	3.15×10^{-3}	1.75×10^{-4}	1.41×10^{-4}

- **Expected:** rect and orthogonalised follow the theory; Gauss×sinc is worse because of its self-interference; the equaliser should repair it.
- **Observed:** rect and orthogonalised sit on the theory curve within the statistical scatter. Gauss×sinc without equalisation falls away: it is 1.2× the theory at 4 dB, 1.9× at 8 dB and about 23× at 12 dB (3.15×10^{-3} against 1.39×10^{-4}), so the curve flattens instead of dropping. With MMSE equalisation it returns to within about 10 % of theory at 6–10 dB and about 26 % above it at 12 dB.
- **A simple check of the loss.** Treat the self-interference as extra Gaussian noise of variance $\varepsilon^2 = 0.1108^2$ on top of the channel noise $1/(E_s/N_0)$, and read the theory curve at the resulting lower SNR. This predicts 2.8×10^{-3} at 12 dB, 7.7×10^{-3} at 10 dB and 1.9×10^{-2} at 8 dB, against the measured 3.15×10^{-3}, 7.4×10^{-3}, 1.79×10^{-2}: within about 10 %.
- **A surprise at low SNR.** Below 4 dB the MMSE version is slightly *worse* than no equalisation (1.55×10^{-1} against 1.46×10^{-1} at 0 dB). This is not a failure of the equaliser: MMSE shrinks its estimates towards zero when noise is strong, while the demapper's decision thresholds are fixed. In a plain AWGN test with no pulse at all, shrinking the received symbol by the same factor $1/(1 + N_0/E_s)$ already raises the BER at 0 dB from 0.141 to 0.152.

4.5 Constellations at 12 dB

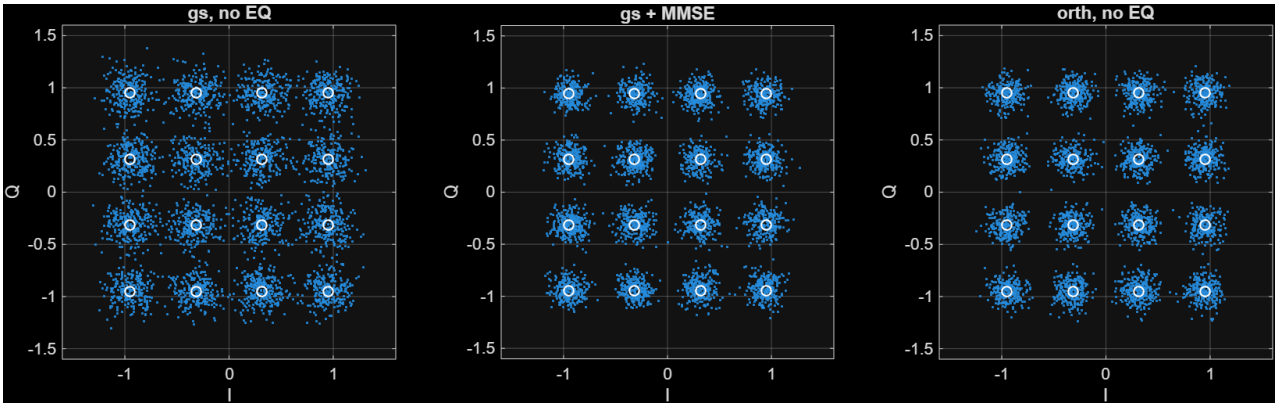


Figure 5: Received symbols at $E_b/N_0 = 12\,\mathrm{dB}$ (8 frames): Gauss×sinc without equaliser, with MMSE, and orthogonalised without equaliser.

- **Expected and observed:** the Gauss×sinc clouds are wider and ragged (noise plus neighbours); after MMSE they are as compact as the orthogonalised clouds, which need no equaliser.

4.6 Noise-free floor

Table 3: Noise-free BER over 500 frames (1.02 Mbit).

Case	bit errors	BER
Rect	0	0
Gauss×sinc, no equaliser	11	1.07×10^{-5}
Gauss×sinc, zero-forcing	0	0
Orthogonalised, no equaliser	0	0

- **Expected:** even without noise, the non-orthogonal pulse can cause errors; the equaliser should remove them.
- **Observed:** 11 errors in over a million bits for Gauss×sinc: the self-interference alone ($\varepsilon = 0.11$) rarely pushes a symbol across a decision boundary (the distance is 0.32). Zero-forcing removes them. The orthogonalised pulse, with $\varepsilon = 0.0149$, has none.
- Takeaway: by itself the self-interference is tolerable; the damage comes when it *adds to* the noise, as in the table above.

4.7 The operator H and its eigenvalues

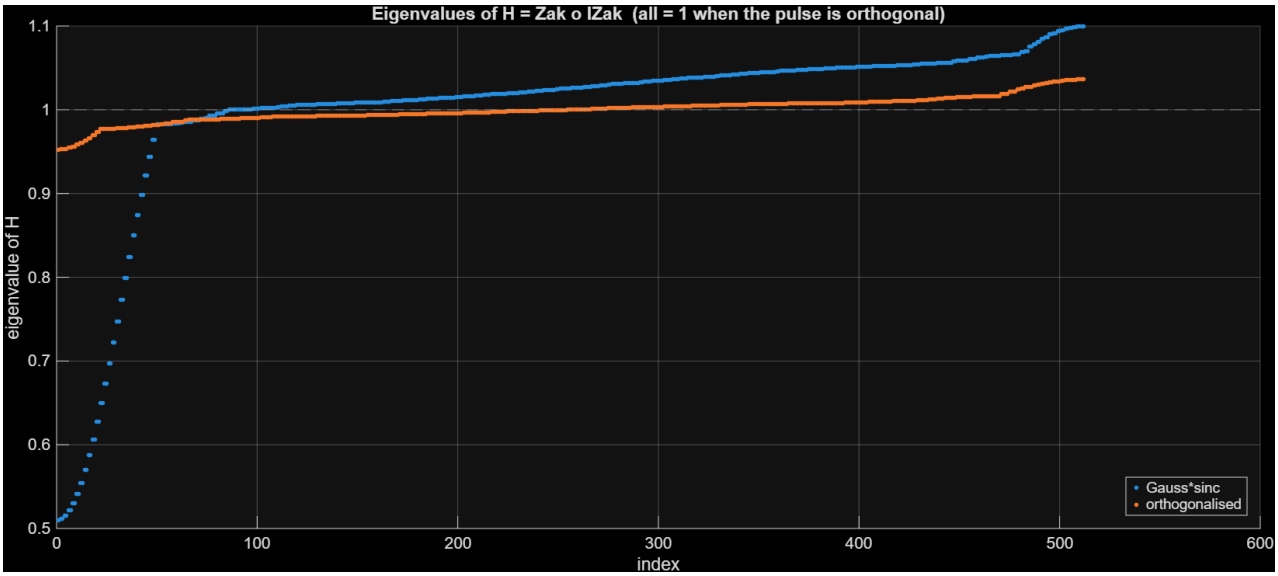


Figure 6: Sorted eigenvalues of H (512×512) for the Gauss×sinc and the orthogonalised pulse. All eigenvalues $= 1$ would mean $H = I$.

- **Expected:** an orthogonal pulse has all eigenvalues at 1; Gauss×sinc spreads them.
- **Observed:** the orthogonalised pulse has eigenvalues between 0.9525 and 1.0362 (mean of $1/\lambda$: 1.0002); Gauss×sinc between 0.5101 and 1.0996 (mean of $1/\lambda$: 1.0200), with a tail of about 50 eigenvalues well below 1 (49 are below 0.97). The matrix model also matches the simulated chain: $|HX - Y| \leq 1.1 \times 10^{-15}$.
- The inverse of a small eigenvalue amplifies noise, so zero-forcing can enhance noise (here by about 2 % on average, $1/0.51 \approx 2\times$ in the worst direction); MMSE avoids that, at the price of the shrinkage seen above.
- Takeaway: H tells in advance how good the pulse is and how to equalise it.

5 Summary

- The full chain works: bit-exact without noise, and the BER follows the 16QAM theory for the rectangular and the orthogonalised pulse.
- Gauss×sinc has about 23× the theoretical BER at 12 dB because its self-interference ($\varepsilon = 0.11$) adds to the noise; treating it as extra noise predicts the loss within about 10 %.
- MMSE equalisation with the known H restores it to within about 5–26 % of theory from 4 to 12 dB, but is slightly worse than nothing at very low SNR.
- The orthogonalised pulse needs no equaliser ($\varepsilon = 0.015$), at the longer pulse cost found in Report 7.
- Speech quality tracks the BER: unusable at 4 dB, glitchy at 8 dB, nearly clean at 12 dB and identical to the PCM source (42.1 dB SQNR) at 16 dB.

Appendix: details and assumptions

- **16QAM mapper.** The mapper was implemented from the structure of TS 38.211 clause 5.1.4, $d = \frac{1}{\sqrt{10}}\big[(1-2b_0)(2-(1-2b_2)) + j(1-2b_1)(2-(1-2b_3))\big]$, which we could not check against the specification text here (the formula is shown only as an image on the sites we could reach).

The Gray property and the agreement with the textbook BER support it, but they do not prove it identical to the standard; the printed constellation should be compared with the standard's table. The script uses MATLAB's `nrSymbolModulate` for a cross-check only if the 5G Toolbox is installed; no such cross-check was printed.

- **Slide example.** The slides' 64QAM example 101001 is given as $3 - j5$ on the transmitter slide (p. 14) and as $5 + 3j$ on the modem-IC slide. The 64QAM formula of clause 5.1.5, as we recall it, gives $-5 + 1j$ (before the $1/\sqrt{42}$ scaling). The two slide values also disagree with each other. The script's printed text says "slide p14 shows $5 + 3j$", which refers to the second slide, not to p. 14.
- **Payload.** The "speech" is synthetic (a sum of 21 harmonics with gliding pitch and slow amplitude modulation, scaled to peak 0.9), not a recording. The 8-bit PCM is uniform between -1 and 1 (step 2/255), MSB first. SQNR is computed after removing the mean of the original.
- **Noise model.** Complex AWGN added to the transmitted samples, variance $\sigma^2 = 1/(N\,E_s/N_0)$ per sample, with the unit-energy pulse and the $1/N$ scaling of the IZak (Report 3). No fading, no Doppler, no synchronisation error and no channel estimation: H is known exactly.
- **MMSE.** $W = (H + E_s^{-1} I)^{-1}$ is derived for noise covariance $E_s^{-1} H$ and unit symbol energy; H is Hermitian (to 10^{-16}), so the inverse is well defined. The equaliser is built for the Gauss×sinc H only.
- The BER points stop at 200 errors, hence the ±7 % scatter. At low SNR the MMSE-versus-none ordering (1.55 against 1.46×10^{-1} at 0 dB) is outside that scatter, and the plain-AWGN test supports the shrinkage explanation.
- The "treat interference as noise" check uses $\varepsilon = \|H - I\|_F / \sqrt{K} = 0.1108$ for the Gauss×sinc pulse and the theory formula above; it is a rough check, not an exact model, since the interference is not Gaussian.
- The number of eigenvalues below 0.97 (49 for Gauss×sinc, 19 for the orthogonalised pulse) is from an independent NumPy replica, as is the plain-AWGN shrinkage test; all other numbers are those printed by `S8_full_chain.m`.
- The PCM figure of the script (original against 8-bit PCM) is omitted; its content is summarised by the 42.12 dB SQNR.
- The figures come from `S8_full_chain.m` (MATLAB dark theme); the four constellation panels of Fig. 2 were cropped to remove empty space.